

# **University Institute of Engineering**

## **Department of Computer Science & Engineering**

### **COMPUTER ORGANIZATION & ARCHITECTURE**

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## COMPARATIVE STUDY OF HARDWIRED AND MICRO-PROGRAMMED CONTROL UNIT

- To execute an instruction, there are two types of control units: Hardwired Control unit and Micro-programmed control unit.
- Hardwired control units are generally faster than microprogrammed designs. In hardwired control, we saw how all the control signals required inside the CPU can be generated using a state counter and a Programmable Logic Array (PLA) circuit.
- A microprogrammed control unit is a relatively simple logic circuit that is capable of sequencing through microinstructions and generating control signals to execute each microinstruction.
- The main difference between Hardwired and Microprogrammed Control Unit is that a Hardwired Control Unit is a sequential circuit that generates control signals while a Microprogrammed Control Unit is a unit with microinstructions in the control memory to generate control signals.

Following are the further differences in tabular form.

| <b>Hardwired Control Unit</b>                                                                                                                              | <b>Microprogrammed Control Unit</b>                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| With the help of a hardware circuit, we can implement the hardwired control unit. In other words, we can say that it is a circuitry approach.              | While with the help of programming, we can implement the micro-programmed control unit.                                                                      |
| The hardwired control unit uses the logic circuit so that it can generate the control signals, which are required for the processor.                       | The micro-programmed CU uses microinstruction so that it can generate the control signals. Usually, control memory is used to store these microinstructions. |
| In this CU, the control signals are going to be generated in the form of hard wired. That's why it is very difficult to modify the hardwired control unit. | It is very easy to modify the micro-programmed control unit because the modifications are going to be performed only at the instruction level.               |

| <b>Hardwired Control Unit</b>                                                                                                                                                | <b>Microprogrammed Control Unit</b>                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The complex instructions cannot be handled by a hardwired control unit because when we design a circuit for this instruction, it will become complex.                        | The micro-programmed control unit is able to handle the complex instructions.                                                                                                  |
| In the form of logic gates, everything has to be realized in the hardwired control unit. That's why this CU is more costly as compared to the micro-programmed control unit. | The micro-programmed control unit is less costly as compared to the hardwired CU because this control unit only requires the microinstruction to generate the control signals. |
| Because of the hardware implementation, the hardwired control unit is able to use a limited number of instructions.                                                          | The micro-programmed control unit is able to generate control signals for many instructions.                                                                                   |
| Difficult to modify as the control signals that need to be generated are hardwired                                                                                           | Easy to modify as the modification need to be done only at the instruction level                                                                                               |

| <b>Hardwired Control Unit</b>                                                   | <b>Microprogrammed Control Unit</b>                                                                     |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| More costlier as everything has to be realized in terms of logic gates          | Less costlier than hardwired control as only micro instructions are used for generating control signals |
| Only limited number of instructions are used due to the hardware implementation | Control signals for many instructions can be generated                                                  |
| Used in computer that makes use of Reduced Instruction Set Computers (RISC)     | Used in computer that makes use of Complex Instruction Set Computers (CISC)                             |

## **Difference Between Hardwired and Microprogrammed Control Unit**

### **Definition**

Hardwired Control Unit is a unit that uses combinational logic units, featuring a finite number of gates that can generate specific results based on the instructions that were used to invoke those responses. Microprogrammed Control Unit is a unit that contains microinstructions in the control memory to produce control signals.

### **Speed**

The speed of operations in Hardwired Control Unit is fast. The speed of operations in Microprogrammed Control Unit is slow because it requires frequent memory accesses.

## **Difference Between Hardwired and Microprogrammed Control Unit**

### **Modification**

To do modifications in a Hardwired Control Unit, the entire unit should be redesigned. In Microprogrammed Control Unit, modifications can be implemented by changing the microinstructions in the control memory. Therefore, Microprogrammed Control Unit is more flexible.

### **Cost**

Furthermore, Hardwired Control Unit are more costly to implement than a Microprogrammed Control Unit.

### **Handling Complex Instructions**

Also, it is difficult for Hardwired Control Unit to handle complex instructions, but is easier for the Microprogrammed Control Unit to handle complex instructions.

## **Difference Between Hardwired and Microprogrammed Control Unit**

### **Instruction Decoding**

Moreover, it is difficult to perform instruction decoding in Hardwired Control Unit than in Microprogrammed Control Unit.

### **Instruction set Size**

In addition to the above differences, the Hardwired Control Unit uses a small instruction set while the Microprogrammed Control Unit uses a large instruction set.

### **Control Memory**

Also, there is no control memory usage in Hardwired Control Unit but, on the other hand, Microprogrammed Control Unit uses control memory.

## **Difference Between Hardwired and Microprogrammed Control Unit**

### **Applications**

Considering the applications, the Hardwired Control Unit is used in processors that use a simple instruction set known as the Reduced Instruction Set Computers (RISC). Microprogrammed Control Unit is used in processors based on a complex instruction set known as Complex Instruction Set Computer (CISC).

## References

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